

FINAL PROJECT REPORT

Analyzing the connection between COVID-19 and the Opioid Epidemic

ABSTRACT:

This study analyzed the potential relationship between the COVID-19 pandemic and opioid overdose deaths. The significance of our research is that by understanding whether or not these two public health crises are related, public health experts can determine how to allocate harm reduction resources better and address any gaps in COVID19 related public health measures (such as masking and vaccination rates).

To conduct our analysis of these two public health crises, we applied data cleaning techniques first to prepare our datasets appropriately using the k-means clustering algorithm and then looked at descriptive statistics, results of the Pearson correlation test, conducted SLR regression modeling, plotted critical data on time plots and scatter plots, and also applied a multilayer perceptron deep learning model to our dataset. With respect to COVID-19 case rates compared to opioid overdose deaths, our findings indicate a weak correlation between these two variables.

INTRODUCTION:

We are interested in the following question: What relationship exists between opioid overdose deaths and COVID-19 cases and deaths? Our problem is understanding how COVID-19 case rates in some of the hardest hit areas in the United States and opioid overdose deaths are related. Regardless of the presence of a relationship between these two public health crises, we will plot this data considering demographic and socioeconomic status.

The potential connection between the rate of COVID-19 cases and the frequency of opioid overdose deaths across the country can help public health experts allocate appropriate resources in communities where this relationship occurs and these experts can also look into the inequities presented by COVID-19 and access to healthcare resources to prevent a drug overdose. This problem is interesting since the COVID-19 situation is constantly unfolding as is the opioid epidemic in the United States, so being able to explore this problem means we have plenty of datasets and periods of time to analyze.

CHALLENGES:

1. Inconsistent formatting and columns between state versus county-level data for the COVID-19 dataset and the drug overdose dataset.
2. Aligning the timeframes to only examine data from January 1st, 2020 to December 31st, 2020.

DATASET:

The dataset contains the usage of opioids, the number of covid cases in different regions, and medical facilities, a prescription rate year-wise, the number of deaths due to covid, number of deaths by the usage of opium. Datafile underlying causes of death 1999-2019.csv contains information on region-wise usage of drugs and crude rate of the population.

We have data on the unemployment caused due to covid and also per capita during the period. Most of the data is summarised and compared among the regions and causes of death.

There are more datasets contained in the project that are used:

- statesCOVID.csv: Used to compare the state-wise covid cases.
- VSRR_Provisional_Drug_Overdose_Death_Counts.csv

- Unemployment.csv: It defines the unemployment data in all states

These datasets have been summarized and can be studied further and the above files were the most commonly utilized file.

METHODOLOGY

1. The first dataset we chose to examine was the U.S. Census race data because we wanted to examine if race was at all correlated to our examination of COVID-19 cases and deaths alongside opioid overdose deaths.
2. The second dataset we chose to examine was the U.S. Census unemployment data since we wanted to see if the unemployment rate was at all correlated with both COVID cases and deaths along with opioid deaths.
3. The third dataset we chose to examine was sourced from the NYTimes state COVID-19 dataset and we chose to examine this dataset since the drug overdose/opioid overdose death dataset was scoped at the state level, so in order to handle scoping issues well, we chose to work at the state level.
4. The fourth dataset we chose was sourced from the VSRR Provisional Drug Overdose Death Counts dataset and we decided to use this dataset since this was the only dataset that had the most recent drug overdose death data up until 2020.
5. The key tools we used were Python3 via JupyterNotebooks Anaconda and libraries from Python (numpy, pandas, matplotlib.pyplot, seaborn, statsmodels.api, scipy.stats, pasty dmatrices, and sklearn clustering, model selection, linear_model, and metrics).
6. The main tasks we carried out were data cleaning to create a larger data frame for analysis purposes, computing descriptive statistics for each variable in the larger data frame, correlation analysis between variables, SLR modeling prior to plotting certain variables, and applying a rudimentary deep learning model.
7. All the datasets needed to be cleaned and preprocessed in some way for this research. The census population and race data had non-descript column names as the column headers and the first row as an actual

descriptive set of column headers. The first row replaced the headers for this data frame, as well as the names were cleaned up by removing the '!' characters before and after each header.

8. The unemployment dataset filtered out all columns except those in the 2020 timeframe, as well as stripping any commas for the numeric values. The COVID-19 data was also filtered to just 2020.
9. Many changes were needed for the opioid data. Unneeded columns were removed. Because the year and month columns were separate in this dataset originally, they were changed and formatted into a singular date column for easier plotting using `pd.to_datetime()`. This dataset also featured all the possible causes of drug overdoses, so all except opioid overdose deaths were filtered out.
10. The "Data Value" column, which held the death count, was changed into the float data type. Floats were favored over integers because of the data imputations for this dataset.
11. The data for many of the datasets needed to be aggregated to a state level and some datasets were formatted for use in a time plot. For the COVID-19 data frame, the totals for 2020 by the state were summed using `pd.Grouper` for both cases and deaths.
12. The COVID-19 by-month and by-state data frame start off as a copy of the data frame with filtered-out states. To obtain the number of cases and deaths per month for each state, the first step was to go through and create a data frame for each state with both COVID-19 and opioid data. Then each data frame aggregated the total number of cases/deaths per month for each state. The totals are cumulative, so the max was grabbed for each month using `pd.Grouper()` rather than the sum. Each state's own individual data frame was then appended back together. This time plot no longer worked with the cases and deaths on each day, so those columns were removed. In addition, the mm/dd/yyyy format was changed to just the month, and duplicates were removed with the days being irrelevant.
13. The date, state, cases, and death attributes were converted to numpy arrays to iterate through. Then in a for loop for the length of the arrays, actual death and cases per month were calculated.
14. A separate data frame was created for COVID-19 totals across all states over time as well, where the data was aggregated using `pd.Grouper()` by month.

15. K-means clustering was used to fill in the data and make an estimate of the number of opioid overdose deaths for states without opioid death data. This was done using the state population, the number of total drug overdose deaths, and the unemployment rate attributes for 2020 only.
16. The total state population needed to be obtained from the census dataset and aggregated. Those attributes were used to cluster the states together into seven clusters. From those clusters, the average proportion of opioid deaths to total drug overdose deaths was taken for each cluster. This average was then applied to the drug overdose death total for each state to make an estimate of the number of opioid overdose deaths.
17. The census race dataset was cleaned to get the breakdown of different races by the state for 2020. There were several categories for mixed races in the original dataset which brought it to 73 total columns.
18. The last step before the analysis was to append all the cleaned and filtered state data into one large data frame called `df_analysis`.
19. This includes opioid overdose data, COVID-19 deaths, and cases both the raw numbers and estimates and per capita, unemployment and labor data, census race data, and total population. This was done for the purpose of making correlation analysis easier.
20. The correlation analysis itself used the built-in panda's data frame function `.corr()`, which outputted a Pearson correlation coefficient for any two variables in the data frame.
21. There were several plots created with the data that we obtained and cleaned. A time plot was created with opioid deaths and COVID-19 deaths across 2020 for each state with available data in both those attributes as well as a plot for all the states combined.
22. The clustering estimates were not used for any of these plots. The next two plots were totals across the US for opioid deaths over time, COVID-19 deaths, and COVID-19 cases. These were chosen to potentially visualize any underlying patterns in the data over time.
23. The next analysis and plots used simple linear regression (SLR) modeling to show the relationship between the plotted variables. The plots for SLR were COVID-19 deaths per capita and opioid deaths per capita, COVID-19 cases per capita and opioid deaths per capita, COVID-19 deaths and unemployment rate, COVID-19 cases, and unemployment rate, and opioid deaths and unemployment rate.

24. The plots scatter plots that have the linear regression line of best fit and the 95% confidence interval limits for the line.
25. The three attributes that were tested for patterns were COVID deaths, COVID cases, and opioid deaths. The function used to test the accuracy of the model was `model.evaluate()`.

GRAPHS

	n	mean	median	Q1	Q3	IQR	stdev	skewness	kurtosis	test_stat	p_value
Opioid Deaths	51	14324.90196	10130	3445	20633.5	17188.5	15101.05959	1.679414682	2.967079076	24.22654638	5.49E-06
COVID Cases	51	390283.451	276834	106546.5	517446.5	410900	435018.8269	2.622025443	8.367266503	46.8790188	6.61E-11
COVID Deaths	51	6752.921569	4787	1461.5	7875.5	6414	7952.422023	2.078495298	4.515697491	33.15557611	6.31E-08
Opioid Deaths Per Capita	51	0.002327229966	0.002214353806	0.001210756571	0.00320149961	0.001990743035	0.001394828965	0.9260489184	0.7204875789	8.220397997	0.01640450973
COVID Cases Per Capita	51	0.06237194153	0.06144003671	0.05155397044	0.0753684362	0.02381446576	0.02148343073	-0.1424308906	0.8115902214	1.795203783	0.4075458278
COVID Deaths Per Capita	51	0.000994709706	0.000956471685	0.000701321896	0.001226348116	0.000525026221	0.000449226221	0.2935688705	-0.3325630952	0.99377432	0.6084216342
Population	51	6499005.51	4505836	1816411	7428391.5	5611980.5	7408022.546	2.652642363	8.493064875	47.4381647	5.00E-11
Unemployment_rate_2020	51	7.368627451	7.3	6.1	8.3	2.2	1.794156091	0.7097692805	0.6374909204	5.513708239	0.0634911902
White	51	4005436.725	3007874	1373700.5	5121600	3747899.5	3686324.558	1.612154104	2.559348143	22.2615902	1.47E-05
Black or African American	51	805964.7059	362417	63990.5	1336370.5	1272380	966965.4949	1.4143327	1.257672034	16.00288514	0.0003349790477
American Indian and Alaska Native	51	73081.07843	31845	19776	71385	51609	109842.9332	3.352185318	13.48886539	61.60970896	4.18E-14
Asian	51	389922.5294	133377	40830	381997.5	341167.5	898639.9214	5.431807635	33.49680837	94.70473816	2.72E-21
Native Hawaiian and Pacific Islander	51	13528.7451	3533	1566	10008.5	8442.5	31479.88335	4.04258722	16.64810847	71.89628258	2.44E-16
Other Race	51	547366.9608	182054	72790	455348	382558	1296114.199	4.982885899	27.8991646	88.15366271	7.21E-20
Two or more races	51	663704.7647	359534	127447.5	693539.5	566092	1124699.756	3.563030295	12.97873367	63.4439806	1.67E-14

Figure 1: Table of all obtained descriptive statistics

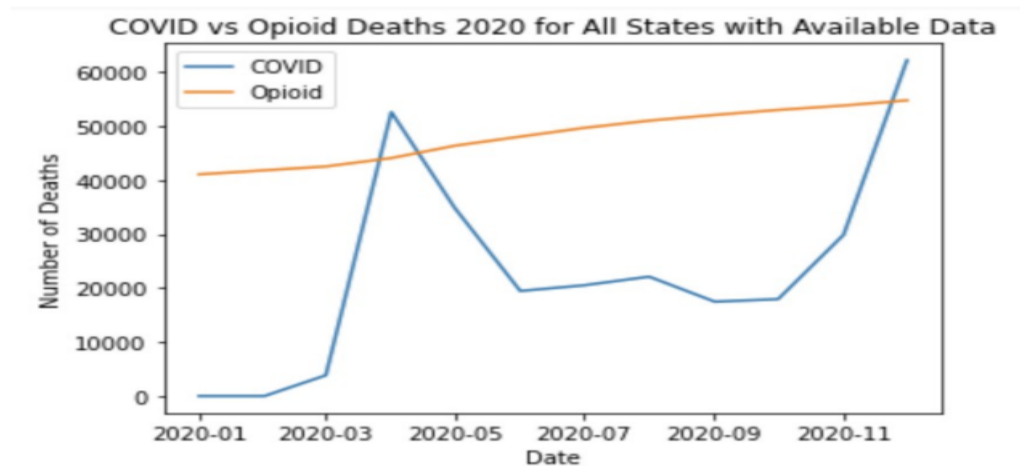


Figure 2: Time plot for COVID-19 deaths and opioid deaths for 2020 for all US

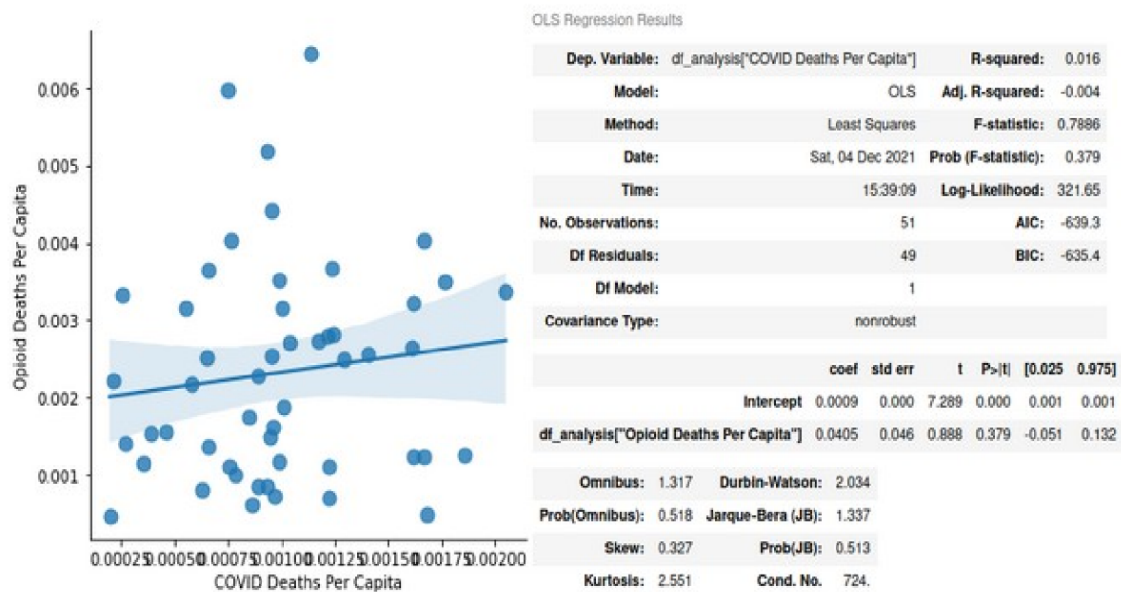


Figure 3: COVID-19 deaths per capita vs opioid deaths per capita

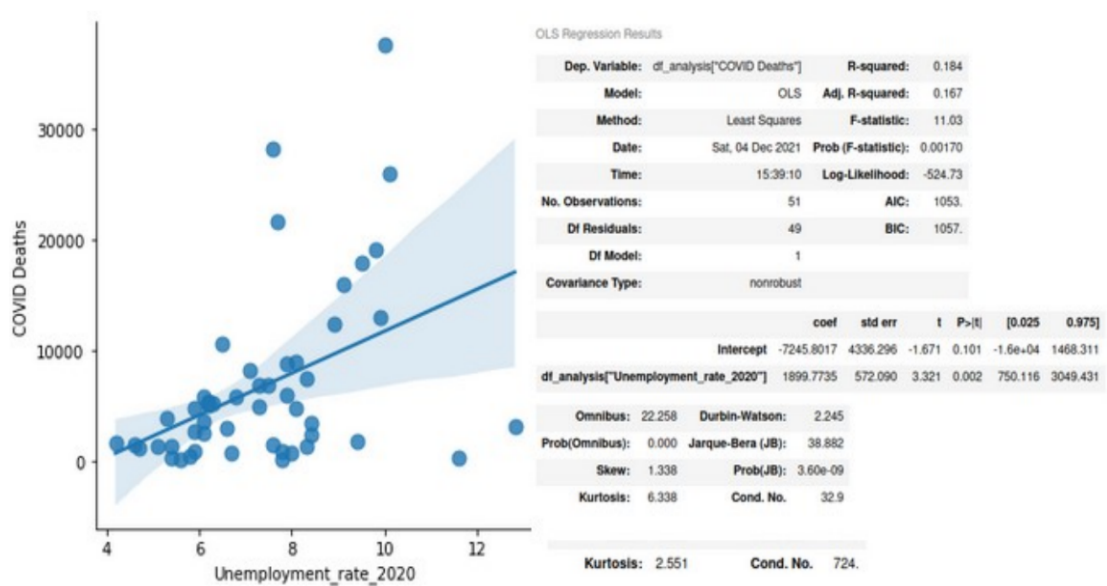


Figure 4: COVID-19 Deaths vs unemployment rate

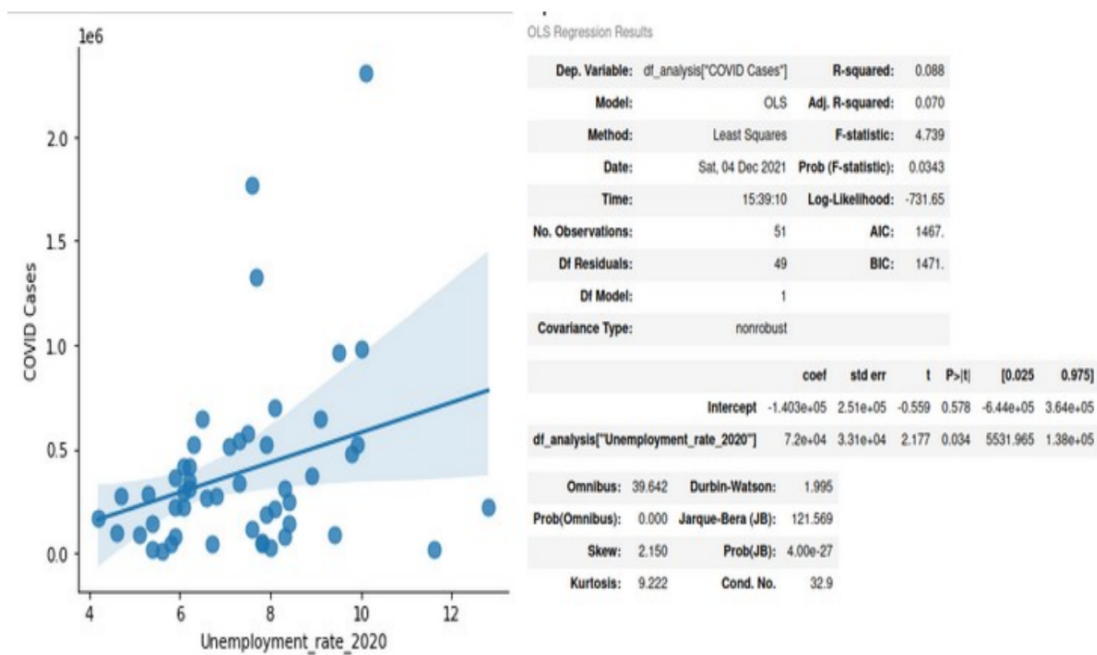


Figure 5: COVID-19 Cases vs unemployment rate

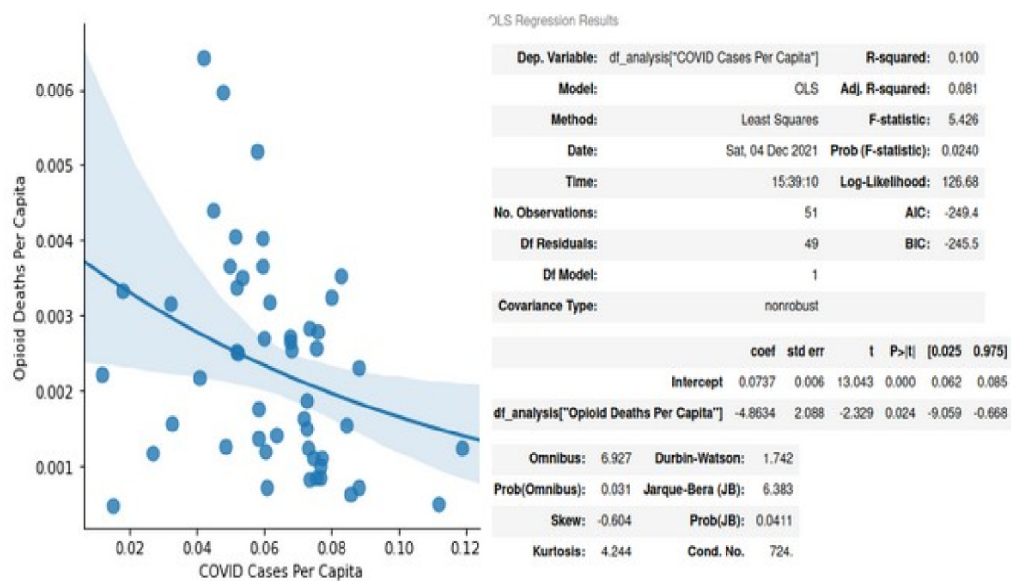


Figure 6: COVID cases per capita vs Opioid deaths per capita

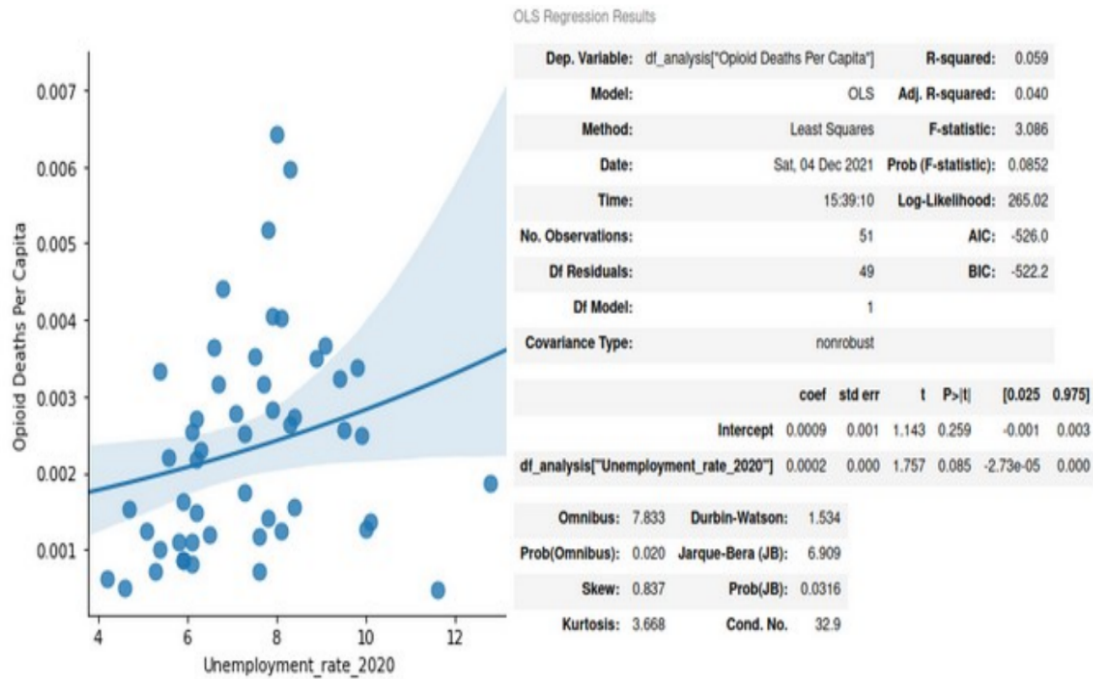


Figure 7: Opioid deaths per capita vs Unemployment rate for 2020

EVALUATIONS

1. Our data was cleaned and filled in using three different methods. For the COVID-19 data, some early months had missing data which were filled in as zeroes, assuming COVID-19 had not been detected in those states.
2. Two states with data had a few missing months of data points in the opioid dataset, which were filled in with the mean value for the state.
3. Lastly, k-means clustering was used to estimate the opioid overdose deaths for states without values using total drug overdose and population data.
4. Our problem in the context of the types of machine learning problems is a regression analysis problem, which means that our approach should be roughly 75% higher accuracy than the predicted median of the training set, the predicted quantile of the dataset, and a predicted constant value from the user.
5. This means that our regression analysis should have a high accuracy compared to the baselines we are dealing with. However, our results showed that our regression analysis models for each set of variables we analyzed did not meet this threshold and we found that most of our R2

was well below the cutoff of 0.7 which is considered a good model with good positive correlation.

6. The first metric used to evaluate our work was descriptive statistics which includes n for the number of samples, mean, median, Q1 representing quartile 1, Q3 representing quartile 3, IQR representing the interquartile range, stdev representing standard deviation, skewness of the data, kurtosis, test statistic value, and p-value (for purpose of hypothesis testing).
7. The evaluation metric derived from dispersion methods such as descriptive statistics helped us to determine the spread of the dataset we are analyzing and the variability present in the dataset.
8. The most crucial findings from the descriptive statistics table is that the average COVID-19 cases per capita is 0.062372, the average COVID-19 deaths per capita is 0.000995, and the average unemployment rate in the U.S. is 7.368627% in the time period 01/01/2020 to 12/31/2020.
9. Kurtosis is high for all non-white alone races since the U.S. has a high population of persons with white racial identity and certain states have high concentrations of certain races.
10. Examining the unemployment rate for 2020 and the opioid deaths in 2020 (not per capita), we found that the correlation coefficient is 0.384841 which suggests a weak positive correlation between these two variables.
11. Examining the unemployment rate for 2020 and the opioid deaths in 2020 per capita, we found that the correlation coefficient is 0.243396 which indicates a weak correlation between the unemployment rate and opioid deaths per capita in 2020.
12. Looking at the COVID-19 cases per capita 2020 and opioid overdose deaths per capita in 2020, the Pearson correlation coefficient is -0.315757 which indicates a weak negative correlation between these two variables.
13. Looking at the COVID deaths (not per capita) in 2020 and opioid overdose deaths (not per capita) from 2020, the Pearson correlation coefficient is 0.728414. While this value may indicate that there is a strong positive correlation between these two variables, it is important to note that these variables are not scaled per capita (by population value per analysis state) and thus have inaccuracies when looking at the raw data itself.

14. Examining the unemployment rate from 2020 and COVID-19 deaths from 2020 (not scaled per capita), the Pearson correlation coefficient is 0.428610, again it is important to note that even though this value indicates a positive correlation between unemployment and COVID-19 deaths, this data is not accounting for COVID-19 deaths per capita.
15. Examining the unemployment rate for 2020 compared to COVID-19 deaths per capita from 2020, the Pearson correlation coefficient is 0.221363 which indicates a weak positive correlation between these two variables accounting for the per capita factor in the COVID-19 deaths variable.
16. The third metric we used to evaluate our findings was a time plot for COVID deaths per capita versus opioid overdose deaths per capita. This plot helps us to understand that while COVID deaths increased variably over the time period between 01/01/2020 to 12/31/2020, opioid overdose deaths remained relatively steady with a slight increase over time.
17. The opioid deaths over time followed some different patterns across the different states, but most states were consistent with the overall US trend of a slight increase through 2020.
18. The fourth metric we used to evaluate our findings was creating SLR models for the scatterplots we created for COVID-19 deaths per capita vs opioid deaths per capita, COVID-19 deaths vs unemployment rate, COVID-19 cases vs unemployment rate, COVID-19 cases per capita vs opioid deaths per capita, and opioid deaths per capita vs unemployment rate.
19. SLR modeling for each of the aforementioned scatterplots helped us to determine if there was a linear relationship between each variable as well as the accuracy of our SLR model.
20. For the scatterplot(Figure 3), looking at COVID-19 deaths per capita versus opioid deaths per capita, the R^2 value is 0.016 and the adjusted R^2 value is - 0.004 which indicates that the input variable (opioid deaths per capita) does not add value to the SLR model for these two variables. This data is difficult to represent with a linear regression model since there is a weak correlation between the two variables and making adjustments to this model is difficult due to the random dispersion of values across this scatterplot.
21. For the scatterplot(Figure 4)looking at COVID-19 deaths (not per capita) and the unemployment rate for 2020, the R^2 value is 0.184 and the

adjusted R2 value is 0.167 which indicates that there is a very weak linear relationship between the two variables and the input variable (unemployment rate) does not add much value to the SLR linear regression model. This data is very difficult to represent with a linear regression model as the pattern in the data is not very well defined between the two variables and there are clusters of data points that would be hard to represent with a linear model.

22. For the scatterplot(Figure 5) looking at COVID-19 cases(not per capita) and the unemployment rate for 2020, the R2 value is 0.088 and the adjusted R2 value is 0.070 which indicates that there is a very weak correlation between these two variables and that the model itself lacks accuracy. This data is quite difficult to represent using a linear regression model or regression analysis in general due to the clustering of data points, weak correlations between the COVID-19 cases and unemployment rates, and a lack of general pattern in the dataset for the sake of modeling.
23. For the scatterplot(Figure 6) looking at COVID-19 cases per capita and opioid deaths per capita, the R2 value is 0.100 and the adjusted R2 value is 0.081 which indicates that there is a very weak correlation between these two variables and that the model itself lacks accuracy since the input variable (opioid deaths per capita) does not add much value to the SLR model. This data is difficult to represent with a linear regression model since the data is spread out in the scatterplot and has no defined pattern to model against.
24. For the scatterplot(Figure 7) looking at opioid deaths per capita and the unemployment rate for 2020, the R2 value is 0.059 and the adjusted R2 value is 0.040 which indicates that there is a very weak correlation between these two variables and that the model itself lacks accuracy based on the adjusted R2 value since the input variable (opioid deaths per capita) does not add much value to the SLR model. Similar to Figs. 5-7, this data is difficult to represent with a linear regression model since the data is spread out in the scatterplot and has no defined pattern to model against.
25. The final metric used to evaluate our data was a deep learning model. This model intended to use regression and predict numeric outputs for opioid deaths, COVID-19 cases, or COVID-19 deaths using a multilayer

perceptron model and all the given attributes for 2020 data. The results from this metric were inaccurate overall.

26. Of the three attributes used for output in separate models, the best result was 9.09% accuracy for predicting COVID Deaths. The other two models had an accuracy so low that they were represented as 0%.

CONCLUSION

1. In this study, we examined the COVID-19 pandemic data on cases and deaths and compared these data to opioid overdose death data from the timeframe January 1st, 2020 to December 31st, 2020.
2. We conducted our analysis using data cleaning techniques to prepare our dataset, and k-means clustering algorithm to fill in missing data in the opioid dataset.
3. Then, we examined the descriptive statistics, Pearson correlation analysis results, SLR modeling results, multilayer perceptron deep learning model results, and a series of time plots and scatter plots.
4. Throughout our analysis, our findings suggest that there is a very weak correlation between opioid overdose deaths compared to COVID-19 deaths, COVID-19 cases, unemployment rate, and racial data accounted for, along with a weak correlation between unemployment rate compared to COVID-19 deaths and cases.
5. Our results indicate that opioid overdose deaths have been consistent over the time period analyzed while COVID-19 deaths were variable during this time period.
6. While the rate of opioid misuse may have been even higher in 2020, prescriptions are being distributed less compared to previous years, and the raw numbers will not reflect a higher rate of misuse.
7. Since the rate of misuse for 2020 data is unfortunately unavailable, this is something we cannot confirm but can hypothesize based on our results. Overall, our findings indicate that there is no discernable pattern between opioid overdose deaths and COVID-19 cases and COVID-19 deaths even after race and unemployment are accounted for.

REFERENCES

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